

DEFORESTATION AS A PUBLIC HEALTH ISSUE

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Introduction: Deforestation & Public Health

- Deforestation = large-scale removal of trees[1]
- Environmental health is tied to ecosystem health[1]



Why Deforestation is a Public Health Concern



Air pollution → worsens respiratory health



Food insecurity → hunger and malnutrition



Water scarcity → unsafe drinking water



Disease spread → malaria, zoonotic diseases



Mental health stress from loss of livelihoods

Deforestation in Zimbabwe

Zimbabwe has lost significant forest cover due to:

- Charcoal & firewood use: 61% of households rely on wood fuel (especially rural)[7]
- Agricultural expansion: Slash-and-burn farming, tobacco curing[7]
- Illegal logging & land clearing[7]
- Urban expansion & mining activities[7]

📊 Data example: Zimbabwe lost 312,000 hectares of tree cover between 2001–2021 (Global Forest Watch) [7]

- Tobacco curing alone consumes over 15 million trees annually[8]

Case Study: Rural Communities and Tobacco Farming

- Tobacco farming = Zimbabwe's top export[8]
- Farmers rely on wood for curing tobacco leaves[8]
- This causes massive deforestation, leading to:

- Soil erosion → reduced food production[1,8]

-Increased flooding in rainy season[1,8]

- Health burden: hunger, malnutrition, vector-borne disease[1,8]

https://youtu.be/dExDpdLOyzI?si=_0itLL9NSd1bfoji

https://youtube.com/shorts/uqlTnTWcnRQ?si=qxN80WYeIAuRK1g_





Public Health Implications in Zimbabwe

- Respiratory illnesses from wood burning[8]
- Waterborne diseases due to river siltation[8]
- Malaria risk from stagnant waters[8]
- Nutrition challenges from biodiversity loss[8]
- Climate vulnerability → droughts, malnutrition[8]



Major Forces Leading To Deforestation

The factors are:

- Agricultural expansion – the largest cause, ~80% of tropical deforestation. Includes palm oil, soy, and cattle ranching, as well as small-scale subsistence farming. [2]
- Logging (legal & illegal) – demand for timber, paper, and wood products. Roads built for logging open forests for further clearing. [2]
- Urbanization & infrastructure – cities, roads, dams, pipelines, and settlements fragment forests. [2]
- Mining & resource extraction – gold, oil, and minerals require clearing large areas and cause pollution.[2]
- Fuelwood & charcoal – in many developing regions (e.g., Nigeria), people rely on wood as their main energy source. [2]



WHY IS DEFORESTATION DIFFICULT TO CONTROL?

- Economic dependence: Governments and communities rely on forests for income and exports[2]
- Poverty: Families need fuel and farmland to survive — alternatives are often too costly[2]
- Global demand: Wealthy countries consume cheap commodities (beef, palm oil, gold) that fuel forest loss[3]
- Weak governance: Corruption, poor law enforcement, and unclear land ownership laws[2,3]
- Cultural practices: Shifting cultivation or charcoal use are long-standing traditions[3]





ECOLOGICAL CONSEQUENCES

Climate change: Forests absorb CO_2. Cutting them increases greenhouse gases and global warming. [1]

Biodiversity loss: 80% of terrestrial species live in forests. Habitat destruction = extinction and loss of medicines. [1,3]

Soil & water disruption: Deforestation causes erosion, flooding, drought, and desertification. [3]

Feedback loops: Loss of forest worsens climate, which further stresses ecosystems. [1]

Consequences of Deforestation To Society [3]

Health risks: Disease spread and poor air quality

Food & water insecurity: Decline in soil fertility → poor crops; altered rainfall → droughts/floods

Economic instability: Short-term profits vs. long-term costs like disasters, health crises, migration, and conflicts.

Increased disaster recovery and poverty





Deforestation vs. U.S. Negative Health Behaviors

🌳 Deforestation[3]

- Driven by logging, farming, mining
- Short-term economic gains
- Environmental harm: climate change, biodiversity loss, soil erosion
- Societal harm: food insecurity, displacement, disasters
- Preventable through sustainable practices

🏥 U.S. Negative Health Behaviors[3]

- Driven by lifestyle, culture, stress
- Short-term pleasure/convenience
- Health harm: obesity, diabetes, heart disease, cancer
- Societal harm: high healthcare costs, reduced productivity
- Preventable through education & healthier choices

➡️ Common Ground: Both provide short-term benefits but create long-term damage to society, requiring prevention and sustainable solutions. [3]

Deforestation in Nigeria

- Nigeria has one of the highest deforestation rates worldwide (~3.5% forest loss annually) [4,5]

Main drivers:

- Fuelwood & charcoal (about 68% of households rely on firewood for cooking) [4]
- Agricultural expansion (slash-and-burn, cocoa farming) [4]
- Logging (legal + illegal timber trade) [4]
- Urbanization & infrastructure growth [4]

Consequences:

- North: desertification advancing southward. [5]
- South: mangrove destruction, oil spill damage, flooding. [5]
- Public health: indoor air pollution, child malnutrition, water scarcity. [4]



Public Health Implications in Nigeria

- **Air pollution:** 68% of households cook with firewood/charcoal → indoor smoke causes pneumonia in children, asthma, COPD in women. [4,5]
- **Diseases:** Forest loss increases risk of malaria, Lassa fever, and other zoonotic spillovers. [4,5]
- **Food insecurity:** Soil degradation + loss of farmland reduce harvests → child malnutrition and stunting. [4]
- **Water insecurity:** Desertification in the north → drought; flooding in the south contaminates water → diarrheal diseases, cholera. [4,5]
- **Conflict & displacement:** Scarcity of land and water drives farmer–herder clashes, migration, and mental health stress. [4]

Nigeria’s Forest Emergency: A Visual Reality



Source: DD India News (2023)

Public Health Interventions & Solutions

- Stronger forest protection laws[2]
- Promote renewable energy (solar, biogas, stoves) [2]
- Community health education[2]
- Agroforestry & sustainable farming[2]
- Research & health surveillance[2]



BIBLICAL PERSPECTIVE

Stewardship of Creation:

Genesis 2:15 – "The Lord God took the man and put him in the garden of Eden to work it and keep it."

God Owns the Earth:

Psalm 24:1 – "The earth is the Lord's, and everything in it."

Justice & the Poor:

Proverbs 14:31 – "Whoever oppresses a poor man insults his Maker."

Creation Care (Hessel & Hessel, 2021)

Protecting forests is a biblical act of stewardship and justice. [6]



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